

From Care to Surveillance:

How Big Data Analytics Challenges Policymaking in Amsterdam's Top600

Yuning Yao (S4244435)

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Leiden University

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Abstract

Data-driven policymaking is increasingly promoted as a means to enhance efficiency and evidence-based decision-making in the public sector, particularly in public safety. This paper critically examines these claims through the case of Amsterdam's Top600 programme, a person-oriented intervention targeting individuals associated with high-impact crime through cross-agency data integration.

Drawing on the frameworks of Data Justice and Contextual Integrity, and situating the analysis within predictive policing literature and the regulatory horizon of the EU Artificial Intelligence Act, the paper analyses how data integration reshapes policymaking processes in practice. The analysis shows that the Top600 generates self-reinforcing feedback loops: intensified surveillance produces additional recorded data, which stabilises risk classifications without demonstrable improvements in recidivism outcomes. Moreover, the programme produces algorithmic stigma by transforming probabilistic risk assessments into durable administrative identities and collapsing contextual boundaries between care and enforcement.

Based on these findings, the paper argues that the core challenges of data-driven policymaking lie not primarily in predictive accuracy, but in governance design. To address these challenges, a technical governance-oriented redesign is proposed, emphasising contextual data separation, role-based access control, temporal limits on classification, and independent auditability. The paper concludes that without such safeguards, data-driven public safety programmes risk undermining effectiveness, legitimacy, and compliance with emerging European AI regulation.

1. Introduction

Data-driven policymaking is increasingly promoted as a pragmatic response to the growing complexity of contemporary governance. Across the European public sector, advances in data infrastructure, interoperability, and analytics are expected to improve the timeliness, targeting, and efficiency of public interventions, while strengthening claims to evidence-based decision-making. In public safety policy, these promises are particularly appealing: if risks can be identified earlier and resources deployed more precisely, governments may be able to prevent harm and reduce recidivism without expanding budgets. At the same time, reliance on integrated data systems raises a central concern for public administration: when policymaking becomes dependent on data infrastructures, the categories and information flows embedded in those systems begin to shape how problems are defined, which populations are prioritised, and how interventions are justified.

This paper examines how big data analytics challenge policymaking processes through the case of Amsterdam’s Top600 programme, a *persoonsgerichte aanpak* (person-oriented approach) targeting individuals linked to high-impact crime. The programme concentrates enforcement and coordinated support on a relatively small group of repeat offenders who, according to the policy rationale, account for a disproportionate share of serious urban crime. Organisationally, the Top600 relies on extensive cross-agency coordination among police, prosecution, probation, municipal services, and particularly in youth-focused variants education and youth care actors. Technically, it operates through a data integration mechanism that compiles individual profiles to support monitoring, prioritisation, and intervention planning (Gemeente Amsterdam, 2022). Rather than using data merely as an input into decision-making, the Top600 restructures decision-making around a unified data infrastructure that connects institutions traditionally separated by distinct mandates and professional norms.

The programme illustrates why “evidence-based” policymaking is not necessarily “justice-based.” While integration is often justified as enabling a balance between enforcement and care, the same data infrastructures that facilitate coordination can also enable intensified surveillance. Placement on a high-risk list can function as an operational signal for street-level actors, legitimising more frequent stops, stricter enforcement of minor infractions, and more aggressive prosecutorial responses. In policing contexts where discretion is central, such practices can generate additional recorded incidents that appear to confirm the original risk classification. These dynamics are commonly described as feedback loops, in which system outputs (targeting) shape future inputs (recorded data), thereby stabilising and amplifying classifications over time (Lum & Isaac, 2016; van Brakel, 2016). From a policymaking perspective, this is consequential because data becomes an endogenous product of governance intensity rather than a neutral measure of behaviour.

Beyond epistemic concerns, the Top600 also raises questions of legitimacy, rights, and contestability. Being labelled a high-risk offender is not merely descriptive; it can become a durable

administrative identity that shapes institutional treatment across domains such as housing, employment, and family life. This process can be understood as *algorithmic stigma*: a form of datafied categorisation that becomes persistent in practice and constrains autonomy by shaping expectations and discretionary decisions (Eubanks, 2018). In the European criminal justice context, civil society reports document how data-driven risk infrastructures are associated with frequent low-level enforcement, heightened surveillance, and spillover harms for families and communities (Fair Trials, 2021). Even where causality is difficult to establish, such accounts point to governance harms that standard outcome metrics often fail to capture.

The importance of these concerns is underscored by empirical evaluation. If data-intensive programmes deliver substantial public safety benefits, their intrusiveness may be defended as proportionate. However, an independent evaluation by the Dutch Research and Documentation Centre (WODC) finds limited evidence that the Top600 produces superior recidivism outcomes compared to comparable offender groups (Beijersbergen et al., 2023). While this does not imply that all components of the approach are ineffective, it challenges the assumption that extensive cross-agency data integration necessarily yields better policy outcomes. It also shifts the justificatory burden towards governance: if outcomes do not improve, what forms of value or harm does the programme produce, and for whom?

To address these questions, the paper adopts three complementary analytical lenses. First, Data Justice foregrounds the distributive consequences of datafied governance, focusing on how individuals are rendered visible and treated through data systems and how these processes reproduce or reshape structural inequalities (Taylor, 2017). Second, Contextual Integrity conceptualises privacy as appropriate information flows within specific social contexts, highlighting the risks of *context collapse* when data collected for care or welfare purposes is repurposed for enforcement and surveillance (Nissenbaum, 2009). The Dutch SyRI judgment illustrates how opaque cross-database risk scoring can be deemed disproportionate and incompatible with fundamental rights (“NJCM c.s. v. State of the Netherlands (SyRI)”, 2020; “SyRI legislation in breach of European Convention on Human Rights”, 2020). Third, the EU Artificial Intelligence Act provides an emerging regulatory benchmark for assessing predictive governance systems in law enforcement, introducing governance requirements that render legacy systems increasingly fragile (“Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)”, 2024).

Building on these lenses, the paper addresses the following research question:

How does cross-agency data integration in the Amsterdam Top600 programme generate feedback loops and algorithmic stigma, and what does this reveal about the challenges of data-driven policymaking under emerging European AI regulation?

The paper advances two core claims. First, the Top600 demonstrates how data integration can destabilise policymaking by producing self-reinforcing feedback loops, in which intensified surveillance generates data that legitimises continued targeting. Second, it shows how integrated

risk classification can produce algorithmic stigma and context collapse, undermining trust in public services and generating long-term distributive harms that are difficult to reconcile with the programme's stated care ambitions. Under the EU AI Act, these dynamics are not merely ethical concerns but increasingly regulatory ones, implying the need for substantial redesign.

The remainder of the paper is structured as follows. Section 2 develops the conceptual framework and reviews relevant literature. Section 3 analyses the Top600 programme through these lenses. Section 4 proposes a technical governance-oriented redesign. Section 5 concludes by discussing the broader implications for data-driven policymaking in public safety and beyond.

2. Conceptual Framework and Literature Review

To analyse how big data analytics challenge policymaking processes, this paper combines three complementary strands of literature: Data Justice, Contextual Integrity, and predictive policing research, with the EU Artificial Intelligence Act serving as a regulatory horizon. Together, these perspectives shift the analysis away from technical performance metrics and towards questions of power, legitimacy, and institutional design. Rather than treating data-driven systems as neutral tools applied to pre-existing policy goals, the framework conceptualises them as governance infrastructures that actively shape problem definitions, population categories, and justificatory logics.

2.1 Data Justice and Automated Inequality

Data Justice provides a normative framework for assessing the social consequences of data-driven governance. Taylor (Taylor, 2017) argues that data justice concerns not only the accuracy or legality of data processing, but how individuals are rendered visible through data, how they are categorised, and how these representations translate into differential treatment. From this perspective, data systems are not merely descriptive; they participate in the production of social order by stabilising categories and legitimising intervention.

In public safety contexts, data justice highlights how administrative and policing datasets are shaped by selective enforcement practices. Arrest records, stop-and-search data, and probation histories reflect institutional attention rather than objective distributions of criminal behaviour. When reused to construct risk profiles, such data can embed historical inequalities into future decision-making, creating path-dependent dynamics in which institutional bias is reproduced through data-driven classification.

Eubanks (Eubanks, 2018) describes this process as *automating inequality*: digital systems scale and formalise forms of social sorting that were previously discretionary, thereby making them more difficult to contest. In criminal justice, risk classifications are particularly consequential because they interact with coercive state power. Once individuals are categorised as

high risk, they may face intensified surveillance, stricter enforcement of minor infractions, and reduced access to rehabilitative opportunities. Crucially, data justice does not presuppose malicious intent; structural injustice can emerge even when systems are designed in pursuit of efficiency or public safety, if their distributive effects disproportionately burden marginalised groups.

Applied to the Top600 programme, a data justice lens shifts attention away from aggregate crime outcomes towards the lived consequences of classification. It raises questions about who is included, how long inclusion persists, and how risk labels shape institutional treatment across domains. If inclusion produces durable disadvantages without corresponding benefits, the legitimacy of the intervention becomes questionable regardless of its stated objectives.

2.2 Contextual Integrity, Purpose Limitation, and Context Collapse

While data justice foregrounds distributive outcomes, Contextual Integrity explains why certain data flows are normatively problematic even when they are legally permissible. Nissenbaum (Nissenbaum, 2009) conceptualises privacy as the preservation of appropriate information flows within specific social contexts, each governed by norms defining who may access information, for what purposes, and under what conditions. Privacy violations occur when information flows across contexts in ways that violate these norms, a process she terms *context collapse*.

In integrated governance systems, context collapse is often justified by appeals to efficiency and coordination. However, Contextual Integrity challenges the assumption that integration is inherently beneficial. When data collected in supportive contexts, such as education or youth care, is repurposed for enforcement or surveillance, the meaning of that data changes. Information provided under conditions of trust and care can become a trigger for punitive intervention.

The Dutch SyRI case illustrates the legal and normative stakes of such practices. SyRI integrated data from multiple government databases to identify individuals at risk of welfare fraud. In 2020, the District Court of The Hague ruled that the framework violated Article 8 of the European Convention on Human Rights due to insufficient transparency, proportionality, and safeguards against discrimination (“NJCM c.s. v. State of the Netherlands (SyRI)”, 2020). Subsequent analyses emphasised that the core issue was not technical non-compliance but the opacity and breadth of cross-context data integration itself (“SyRI legislation in breach of European Convention on Human Rights”, 2020). The judgment has since been widely interpreted as a warning against large-scale, opaque risk profiling in the public sector.

In the context of the Top600, Contextual Integrity highlights how civil and care-related signals are repurposed within a policing-oriented governance framework. When indicators such as school absenteeism or youth care involvement contribute to risk assessments that justify intensified law enforcement attention, individuals may disengage from public services to avoid generating potentially harmful data. From a policymaking perspective, this erosion of trust undermines the preventive and coordinative goals that integration is meant to support.

2.3 Predictive Policing, Feedback Loops, and Algorithmic Stigma

Predictive policing literature offers an operational account of how data-driven classification functions in practice. Predictive policing refers to the use of data analytics to anticipate criminal activity, either by forecasting locations or identifying individuals. While the Top600 is not a fully automated algorithmic system, its logic closely aligns with person-based predictive approaches: individuals are selected, monitored, and prioritised based on aggregated data signals.

A systematic review by Meijer and Wessels (Meijer & Wessels, 2019) finds that evidence for the effectiveness of predictive policing is mixed and highly context-dependent. They note that many widely discussed harms, such as discrimination, stigmatization, and feedback loops, are theoretically plausible but insufficiently evaluated through independent empirical research. This observation underscores the importance of case-based analyses that focus on governance mechanisms rather than outcome metrics alone.

One such mechanism is the data feedback loop. Lum and Isaac (Lum & Isaac, 2016) show that predictive systems trained on police data tend to predict future policing activity rather than underlying crime. Increased attention to certain individuals or areas raises the likelihood that offences will be detected and recorded, reinforcing the original classification in a self-validating cycle. At the individual level, this means that being labelled high risk can itself increase the probability of future recorded incidents, irrespective of behavioural change.

These dynamics contribute to what can be described as algorithmic stigma. Van Brakel (van Brakel, 2016) argues that pre-emptive surveillance renders individuals legible primarily as risks, reshaping institutional relations and discretionary practices. Over time, probabilistic classifications can harden into quasi-permanent bureaucratic identities. Algorithmic stigma thus arises not only from automated outputs, but from sustained institutional reliance on data-driven categories that structure discretion and justify intensified control.

From a policymaking perspective, these insights highlight a core limitation of traditional evaluation methods. Even when aggregate crime indicators remain stable or decline, the burdens of surveillance, enforcement, and stigma may become increasingly concentrated. Systems designed to optimise efficiency may therefore undermine fairness and legitimacy if their internal feedback dynamics are not explicitly addressed.

2.4 The EU Artificial Intelligence Act as a Regulatory Horizon

The EU Artificial Intelligence Act introduces a risk-based regulatory framework that is directly relevant to predictive and integrative governance systems. It distinguishes prohibited practices from high-risk applications and imposes extensive governance obligations on the latter, particularly in law enforcement contexts (“Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)”, 2024). These obligations include requirements related to data quality, transparency, documentation, human oversight, and avenues for explanation and redress.

Although the Top600 predates the AI Act and does not rely on a single identifiable AI model, its functional logic resembles practices that the regulation seeks to constrain. The Act places heightened scrutiny on systems involving individual risk prediction and those with significant potential impacts on fundamental rights. From this perspective, the Top600 appears increasingly regulatorily fragile: even if formally lawful under existing national frameworks, its reliance on broad individual profiling and cross-context data integration may be difficult to reconcile with emerging EU standards.

Treating the AI Act as a regulatory horizon rather than a narrow compliance checklist enables a forward-looking analysis. It shows that data-driven policymaking is challenged not only by ethical critique but also by shifting legal norms that redefine acceptable forms of governance. For policymakers, this implies that legacy systems must be assessed not solely on past justifications, but in light of future regulatory expectations.

3. Case Description and Analysis: The Top600 Programme

This section analyses the Amsterdam Top600 programme as an empirical case of data-driven policymaking in public safety. Rather than treating the programme as a discrete technological artefact, the analysis conceptualises Top600 as a governance arrangement in which organisational coordination, data integration, and discretionary practices are tightly interwoven. The section outlines the programme's policy rationale and institutional design, examines its data infrastructure and operational logic, and discusses findings from independent evaluation. This grounding is necessary to show how feedback loops, algorithmic stigma, and context collapse emerge in practice.

3.1 Policy Origins and Objectives

The Top600 programme was introduced by the municipality of Amsterdam in response to concerns about high-impact crime and repeat offending. Its policy rationale rests on a concentration thesis: a relatively small group of individuals is assumed to account for a disproportionate share of serious offences, including robbery, burglary, and violent crime. By identifying this group and intensifying interventions, policymakers seek to reduce crime while using public resources more efficiently.

From the outset, the programme was framed as a *persoonsgerichte aanpak*, emphasising tailored interventions rather than purely punitive measures. Official communications stress that the Top600 combines enforcement with care, integrating criminal justice responses with social support measures such as housing assistance, debt counselling, education, and youth care (Gemeente Amsterdam, 2022). This framing is central to the programme's legitimacy, as it presents extensive information sharing as a preventive and rehabilitative strategy rather than a solely repressive one.

Institutionally, the programme is coordinated through Amsterdam’s Action Center for Safety and Care (Actiecentrum Veiligheid en Zorg, AcVZ), which functions as a hub connecting police, prosecution, probation services, municipal departments, and, in cases involving minors, education and youth care organisations. Regular case conferences are used to align interventions and monitor progress. While this networked structure reflects broader trends in Dutch safety governance, it also depends on extensive data sharing and stable classification mechanisms, raising governance concerns that extend beyond managerial efficiency.

3.2 Data Infrastructure and Selection Mechanisms

The operational core of the Top600 lies in its data infrastructure. Individuals are selected for inclusion based on a combination of criminal justice data and, in some cases, civil indicators. Police records, prosecution data, probation information, and prior convictions form the backbone of the selection process. For youth-focused variants, additional signals such as school absenteeism, youth care involvement, and social services data may be considered (Gemeente Amsterdam, 2022).

Although selection is not presented as fully automated, data aggregation is central to rendering individuals legible as governance objects. Information from multiple sources is consolidated into individual profiles that enable ongoing monitoring and prioritisation. Once included, individuals are subject to periodic review, yet criteria for inclusion and exit remain only partially transparent to those affected.

This hybrid model, combining aggregated data with discretionary judgment, is significant from a data-driven policymaking perspective. It illustrates that algorithmic effects do not require fully automated decision-making. Even with humans formally “in the loop,” the structure of information can shape discretion by foregrounding certain signals and marginalising others. Risk labels derived from aggregated profiles function as cognitive shortcuts that influence how frontline actors interpret behaviour and decide when to intervene.

Moreover, the integration of civil and criminal justice data blurs traditional boundaries between support and enforcement. Indicators originally collected for care or assistance can become part of a risk profile that legitimises intensified surveillance. This repurposing exemplifies the context collapse identified in the conceptual framework and complicates accountability, as no single institution fully controls how integrated profiles are constructed or used.

3.3 Intervention Logic and Operational Practices

Once individuals are included in the Top600 programme, a range of interventions may be deployed. These include intensified police attention, frequent checks, targeted enforcement of outstanding fines or warrants, and close coordination with probation services. At the same time, care-oriented measures such as housing support, debt restructuring, education or employment guidance, and youth care interventions may be offered or mandated.

In practice, the balance between care and control is variable and contingent on perceived risk and compliance. Nevertheless, inclusion in the programme itself raises the baseline level of institutional attention. Police officers are aware of Top600 status, and this knowledge can influence discretionary decisions during encounters. Minor infractions that might otherwise be resolved informally may lead to formal sanctions, while ambiguous situations are more likely to be interpreted through a lens of suspicion.

These outcomes are not necessarily the result of explicit policy directives. Rather, they emerge from the interaction between data-driven classification and street-level discretion. Once an individual is categorised as high risk, intensified monitoring becomes institutionally rational, even when it increases the likelihood of detecting low-level offences. This interaction forms the basis of self-reinforcing feedback loops, in which surveillance generates data that reinforces classification and legitimises further surveillance.

3.4 Independent Evaluation and Empirical Outcomes

Assessment of the Top600 is informed by an independent evaluation conducted by the Dutch Research and Documentation Centre (WODC). The study compares recidivism outcomes for individuals included in the Top600 with those of comparable offender groups not subjected to the programme, using matched comparison methods (Beijersbergen et al., 2023).

The findings challenge the programme's core assumption. Overall, the evaluation finds limited evidence that participation in the Top600 leads to a statistically significant reduction in recidivism compared to similar offenders outside the programme. While some subgroup effects are observed, the general conclusion is that the intensive, integrated approach does not consistently outperform less intrusive interventions.

These results are consequential for data-driven policymaking. If a programme built on extensive data integration does not deliver superior outcomes, its justification must be reconsidered. The evaluation suggests that the Top600 may be more effective at coordinating institutional responses than at altering behavioural trajectories. In this sense, the programme produces a coherent governance apparatus without clear evidence of improved social outcomes.

3.5 Data Integration and Self-Reinforcing Feedback Loops

Beyond aggregate outcomes, the Top600 illustrates how data integration can actively reshape policymaking processes through self-reinforcing feedback loops. Inclusion in the programme alters the conditions under which behaviour is observed and recorded. Heightened police attention and intensified monitoring increase the likelihood that minor infractions, ambiguous encounters, or technical violations are detected and formally recorded.

These records enter administrative systems as new data points, reinforcing the appearance of persistent risk. This process reflects what predictive policing scholars describe as endogenous data generation. Lum and Isaac (Lum & Isaac, 2016) show that systems trained on police-

generated data tend to predict future policing activity rather than underlying crime. Although the Top600 is not fully automated, the same logic applies: data is produced by intervention rather than merely informing it.

Crucially, these feedback loops can persist even in the absence of improved outcomes. The WODC evaluation indicates that intensified targeting under the Top600 does not consistently reduce recidivism (Beijersbergen et al., 2023). Yet the accumulation of surveillance-generated data can sustain perceptions of ongoing risk, thereby legitimising the programme's continuation. In this way, data integration supports policy persistence through the reproduction of justificatory evidence rather than demonstrable effectiveness.

3.6 Algorithmic Stigma as a Governance Outcome

The Top600 also demonstrates how data-driven classification can produce algorithmic stigma as a structural governance outcome. Inclusion on the list functions as a signal that circulates across institutions, shaping discretionary decision-making by police officers, probation services, and municipal actors. Encounters are more likely to be interpreted through suspicion, tolerance for non-compliance decreases, and punitive responses may escalate.

Over time, risk classifications can become detached from specific behaviours and operate as generalised assessments of character or dangerousness. Van Brakel (van Brakel, 2016) describes this as the transformation of individuals into pre-emptive subjects, governed on the basis of anticipated risk rather than demonstrated action. Algorithmic stigma emerges not from explicit design choices but from the stabilisation of data-driven categories within bureaucratic routines, particularly when inclusion and exit criteria are opaque.

Civil society reports document how such stigma is experienced in practice, including repeated police encounters, difficulties accessing employment or housing, and spillover harms for family members (Fair Trials, 2021). While these accounts do not establish causality on their own, they reveal governance harms that are largely invisible in evaluations focused on aggregate crime metrics. From a Data Justice perspective, algorithmic stigma represents a distributive harm that disproportionately affects already marginalised populations (Taylor, 2017).

3.7 Context Collapse and the Erosion of Trust

Algorithmic stigma is further intensified by the collapse of contextual boundaries inherent in the Top600's data integration model. Information originating in care, education, or welfare contexts can contribute to enforcement-oriented risk assessments, altering the meaning of the data and the expectations under which it was originally provided (Nissenbaum, 2009).

For individuals subject to the programme, this context collapse can erode trust in public institutions. Engagement with supportive services may be perceived as risky if it can indirectly trigger heightened surveillance or enforcement, creating a chilling effect that discourages participation. From a policymaking perspective, this undermines preventive capacities and shifts

the balance further towards coercive intervention.

The SyRI judgment provides a legal analogue for these concerns. In ruling against SyRI, the court emphasised that large-scale, opaque data integration undermines transparency and proportionality, particularly when individuals lack meaningful opportunities to understand or contest risk assessments (“NJCM c.s. v. State of the Netherlands (SyRI)”, 2020; “SyRI legislation in breach of European Convention on Human Rights”, 2020). Although the Top600 operates within a different legal framework, the normative issues are closely aligned.

3.8 From Policy Instrument to Governance Infrastructure

Taken together, the feedback loops, algorithmic stigma, and context collapse observed in the Top600 indicate that data integration transforms policy instruments into governance infrastructures. Integrated data systems do not merely support decision-making; they shape how problems are perceived, how success is measured, and how responsibility is distributed across institutions.

This transformation has two key implications. First, it challenges the assumption that better data necessarily leads to better policy. When data is endogenous to intervention, it may obscure rather than clarify causal relationships. Second, it highlights the limits of incremental reform. Addressing the harms associated with the Top600 requires rethinking the design principles of data integration itself, including purpose limitation, contestability, and the separation of supportive and punitive functions.

These insights motivate the following section, which proposes a technical governance-oriented redesign of the Top600’s information architecture. Rather than optimising predictive accuracy, the proposed approach focuses on mitigating feedback loops, reducing stigma, and restoring contextual boundaries in line with emerging regulatory standards.

4. Proposed Technical Solution

The preceding analysis suggests that the core problems raised by the Top600 programme do not primarily stem from predictive inaccuracy, but from how cross-agency data integration restructures governance: it intensifies surveillance through feedback dynamics, stabilises risk labels, and collapses contextual boundaries. This points to system architecture and information governance as the most leverageable intervention points. This section proposes a governance-oriented technical redesign that operationalises privacy-by-design and contestability-by-design to mitigate feedback loops, reduce algorithmic stigma, and restore contextual separation.

Rather than introducing new predictive models, the proposal redesigns the information infrastructure that supports decision-making. This aligns with algorithmic governance research emphasising that many harms attributed to “AI” arise from institutional embedding and data flows rather than automation alone (Lepri et al., 2018). The objective is therefore not optimisation, but constraint: specifying which information can circulate, under what purposes, and with

what accountability.

4.1 Design Principles

The redesign is guided by four principles derived from the conceptual and empirical analysis.

First, *contextual separation* should be enforced technically, not through organisational promises. Data collected in different institutional contexts (e.g., policing, youth care, education, welfare) should not be freely interchangeable; access must be constrained by role and purpose in line with Contextual Integrity (Nissenbaum, 2009).

Second, *data minimisation and proportionality* should be implemented at the system level. Actors should only see what is strictly necessary for a specific task, reducing the likelihood that aggregated profiles operate as generalised suspicion indicators.

Third, *contestability and temporal limitation* must be built into classification infrastructure. Risk status should be time-bounded, governed by explicit renewal and exit criteria, and visible to both administrators and affected individuals.

Fourth, *auditability and independent oversight* should be treated as core system requirements. The system must support evaluation of both effectiveness and distributive impact, without allowing evaluation data to feed back into operational targeting.

These principles align with governance requirements for high-risk systems under the EU Artificial Intelligence Act, including transparency, human oversight, and risk management (“Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)”, 2024).

4.2 Role-Based Access Control and Contextual Views

The technical core of the redesign is role-based access control (RBAC) combined with fine-grained data tagging. Each data element is assigned metadata specifying its context of origin (e.g., policing, education, youth care), intended purpose, and permitted audience.

Users do not access a unified “risk profile.” Instead, they access *contextual views* tailored to institutional role. Police access is limited to justice-relevant information (e.g., verified convictions, active warrants, court-imposed conditions). They do not access school attendance, youth care histories, or welfare engagement. Conversely, care providers access support-relevant information without exposure to enforcement-oriented labels.

Technically, this can be implemented via attribute-based access control layered on RBAC, where access decisions are evaluated dynamically based on user role, declared purpose, and data attributes. All access requests are logged and periodically reviewed. This directly addresses context collapse by constraining cross-context information flows by design.

From a governance perspective, contextual views reduce two key risks. They limit the circulation of risk labels across domains (reducing stigma spillover) and strengthen accountability by linking access to explicit role and purpose.

4.3 Decoupling Support Triage from Punitive Escalation

To mitigate feedback loops, the redesign separates support-oriented signals from punitive escalation mechanisms. In the current Top600 configuration, social vulnerability indicators can indirectly contribute to enforcement intensity once integrated into a single profile. The redesign implements two technically distinct layers.

The first is a *support triage layer*, accessible only to care-oriented actors. It aggregates indicators such as housing instability, educational disengagement, or debt for the sole purpose of coordinating support. These signals are not translated into enforcement risk scores and cannot trigger punitive action.

The second is a *law enforcement layer*, restricted to a narrowly defined set of justice-related signals. This layer excludes dynamic measures of police contact frequency that are susceptible to feedback effects. Instead, it relies on verifiable legal events and incorporates time decay functions and uncertainty indicators to prevent classification persistence through observational intensity.

This separation reduces the likelihood that intensified surveillance generates self-validating data, while also protecting trust in supportive institutions by ensuring that care does not become a proxy for suspicion.

4.4 Temporal Limits, Exit Criteria, and Contestability

A central driver of algorithmic stigma is classification persistence. To address this, the redesign introduces automated temporal limits and explicit exit criteria.

Risk classifications are assigned a predefined validity period and expire unless renewed under clearly specified conditions. Renewal criteria are grounded in verifiable legal events rather than cumulative contact frequency. The system enforces these rules automatically, reducing the risk of indefinite inclusion through organisational inertia.

In parallel, the redesign provides an explanation interface for affected individuals. Even where full operational transparency is not feasible, individuals should be informed about general inclusion and exit criteria and the duration of classification. This supports contestability and aligns with the AI Act’s emphasis on transparency and redress for high-risk systems (“Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)”, 2024). Technically, these features are straightforward, but governance-relevant: they transform risk status from a quasi-permanent label into a time-bounded administrative measure subject to review.

4.5 Independent Audit Interface and Bias Monitoring

The redesign includes a dedicated audit interface for independent oversight bodies. It provides access to aggregated, anonymised metrics to assess effectiveness and distributive impact without

exposing operationally sensitive individual-level details.

Key indicators include disparities in inclusion rates across neighbourhoods, changes in police contact frequency following inclusion, and outcome measures beyond arrests, such as housing stability or educational engagement where appropriate. Crucially, audit data is insulated from operational decision-making to avoid reintroducing feedback effects through evaluation.

This responds to predictive policing critiques regarding limited independent evaluation of harms (Meijer & Wessels, 2019) and strengthens evidence-based policymaking by enabling systematic scrutiny of whether the system aligns with stated goals and legal constraints.

4.6 Feasibility and Limitations

The proposed redesign is feasible within typical public-sector IT environments. It does not require new data sources or advanced machine learning; it relies on access control, metadata management, and logging capabilities that are widely available. The primary constraints are institutional, including inter-agency agreement on purpose limitation, governance ownership of access rules, and the empowerment of independent oversight.

The redesign also has limits. Technical safeguards cannot eliminate discretionary bias or structural inequality, but they can constrain the conditions under which data integration amplifies harm. The proposal should therefore be understood as a necessary, though not sufficient, condition for more legitimate and just data-driven public safety governance.

5. Conclusion

This paper examined how big data analytics challenge policymaking processes through an in-depth analysis of Amsterdam's Top600 programme. Using the lenses of Data Justice, Contextual Integrity, predictive policing research, and the EU Artificial Intelligence Act, the analysis shows that the central challenges of data-driven policymaking do not arise primarily from technical error or regulatory non-compliance. Rather, they emerge from how data integration reorganises governance itself, reshaping evidentiary standards, redistributing discretionary power, and redefining what counts as legitimate policy success.

A first core insight concerns the epistemic role of data in policymaking. In the Top600, data does not function as an external evidence base against which policy performance can be evaluated. Instead, it becomes endogenous to intervention: intensified surveillance generates additional recorded incidents, which in turn reinforce risk classifications and justify continued targeting. This self-referential dynamic undermines a foundational assumption of evidence-based policymaking, that data can serve as an independent check on policy effectiveness. When evaluation and intervention collapse into the same infrastructure, data risks legitimising policy persistence rather than enabling critical reassessment.

A second insight relates to the distributive consequences of data-driven governance. The

Top600 illustrates how risk classifications can harden into durable administrative identities, producing algorithmic stigma as a structural outcome rather than an unintended side effect. These classifications circulate across institutional domains and shape discretionary practices, concentrating surveillance, enforcement, and social consequences on a limited population. From a Data Justice perspective, such concentration constitutes a distributive harm that is largely invisible to conventional performance indicators focused on aggregate crime rates. Policymaking that relies on these indicators alone therefore risks mistaking administrative coherence for social effectiveness.

Third, the analysis highlights the governance implications of context collapse. By integrating data from policing, welfare, education, and youth care, the Top600 erodes normative expectations about appropriate information flows. Engagement with supportive institutions becomes entangled with enforcement-oriented risk assessment, weakening trust and generating chilling effects that undermine preventive goals. This finding challenges a dominant policy narrative in which integration and coordination are treated as unambiguous goods. In practice, integration without strict contextual boundaries can be counterproductive, particularly in domains where cooperation and trust are essential policy resources.

Taken together, these dynamics reveal a broader transformation in data-driven policymaking: policy instruments increasingly function as governance infrastructures. Integrated data systems do not merely support decision-making; they actively shape how problems are perceived, how responsibility is allocated, and how legitimacy is produced. Once embedded, these infrastructures stabilise particular policy logics and make alternative approaches harder to imagine, let alone implement. This helps explain why data-driven programmes may persist even when empirical evaluations question their effectiveness.

The EU Artificial Intelligence Act sharpens the stakes of this transformation. By foregrounding proportionality, contestability, transparency, and fundamental rights, the Act re-frames data-driven governance as a matter of institutional design rather than technical optimisation. Under this emerging regulatory logic, systems like the Top600 appear increasingly fragile. Compliance cannot be achieved through marginal adjustments or improved documentation alone; it requires rethinking how data integration itself is structured and justified.

In response, this paper proposed a technical governance-oriented redesign. The proposal demonstrates how architectural choices, such as role-based access control, contextual separation, temporal limits on classification, and independent auditability, can operationalise normative principles within technical systems. While such measures cannot eliminate structural inequality or discretionary bias, they can constrain the conditions under which data integration amplifies harm and insulates policy from critique.

More broadly, the Top600 case underscores a central lesson for data-driven policymaking. Big data analytics do not simply enhance governmental capacity; they embed particular assumptions about risk, responsibility, and control into the fabric of governance. Treating data integration as a neutral efficiency tool obscures its political and ethical dimensions. Policy-

makers must therefore confront a more fundamental question: not whether data-driven systems work, but what kinds of governance relations they produce and normalise.

In conclusion, the challenges posed by big data analytics in public safety cannot be resolved by better data alone. They require institutional humility, explicit normative choices, and technical designs that prioritise proportionality and contestability over managerial coherence. The Amsterdam Top600 programme demonstrates both the appeal and the dangers of data-driven governance. Whether similar initiatives can be reconciled with democratic values under emerging AI regulation will depend less on predictive sophistication than on the willingness to redesign the governance logics that data infrastructures embody.

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